

Experiment 07:

A. Passive Reconnaissance through network commands

❖ WHOIS:

```
student@Shivam:~$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2026-09-07T09:22:52Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2028-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-10-01T06:13:28Z <<<
```

Observation:

The **whois github.com** command displays publicly available domain registration information such as the registrar, registration details, name servers, and other domain-related information.

❖ Dig (Domain Information Groper):

```
student@Shivam:~$ dig www.sfit.ac.in

; <<>> DiG 9.18.39-0ubuntu0.22.04.6-Ubuntu <<>> www.sfit.ac.in
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 45419
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:;, udp: 65494
;; QUESTION SECTION:
;www.sfit.ac.in.                IN      A

;; ANSWER SECTION:
www.sfit.ac.in.                7180    IN      CNAME   sfit.ac.in.
sfit.ac.in.                    7180    IN      A       199.79.62.121

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Thu Oct 01 11:49:49 IST 2026
;; MSG SIZE rcvd: 73
```

Observation:

The **dig www.sfit.ac.in** command provides DNS information for the domain. The output shows the DNS response and the IP address associated with the domain.

❖ Traceroute:

```
student@Shivam:~$ traceroute sfit.ac.in
traceroute to sfit.ac.in (199.79.62.121), 30 hops max, 60 byte packets
 1  sophos.sfit.co.in (192.168.7.254)  0.795 ms  0.762 ms  0.743 ms
 2  static-125.131.252.123-tataidc.co.in (123.252.131.125)  5.599 ms  5.580 ms  5.952 ms
 3  * 10.129.10.226 (10.129.10.226)  5.916 ms  5.897 ms
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * if-bundle-34-2.qcore2.esin4-singapore.as6453.net (180.87.36.41)  54.507 ms *
 9  * * *
10  ae-72.a02.sngpsi07.sg.bb.gin.ntt.net (129.250.66.9)  60.694 ms  61.397 ms  60.657 ms
11  * * *
12  * ae-8.r29.osakjp02.jp.bb.gin.ntt.net (129.250.2.67)  124.017 ms ae-13.r35.tokyjp05.j
13  * ae-1.r26.lsanca07.us.bb.gin.ntt.net (129.250.2.176)  244.883 ms  244.865 ms
14  ae-2.a03.lsanca07.us.bb.gin.ntt.net (129.250.3.91)  248.496 ms  238.066 ms  260.110 m
15  ce-3-0-1.a03.lsanca07.us.ce.gin.ntt.net (168.143.228.173)  236.648 ms  251.666 ms  23
16  162-215-195-144.unifiedlayer.com (162.215.195.144)  265.007 ms  255.451 ms  270.194 m
17  69-195-64-111.unifiedlayer.com (69.195.64.111)  265.186 ms 69-195-64-113.unifiedlayer
om (69.195.64.111)  268.346 ms
18  162-144-240-175.unifiedlayer.com (162.144.240.175)  273.730 ms 162-144-240-171.unifie
iedlayer.com (162.144.240.175)  270.779 ms
19  cp-34.webhostbox.net (199.79.62.121)  251.143 ms  237.470 ms  237.016 ms
```

Observation:

The **traceroute sfit.ac.in** command displays the sequence of network hops between the source system and the destination server. It helps identify the route followed by packets through intermediate routers.

❖ Nslookup:

```
student@Shivam:~$ nslookup sfit.ac.in
Server:          127.0.0.53
Address:         127.0.0.53#53

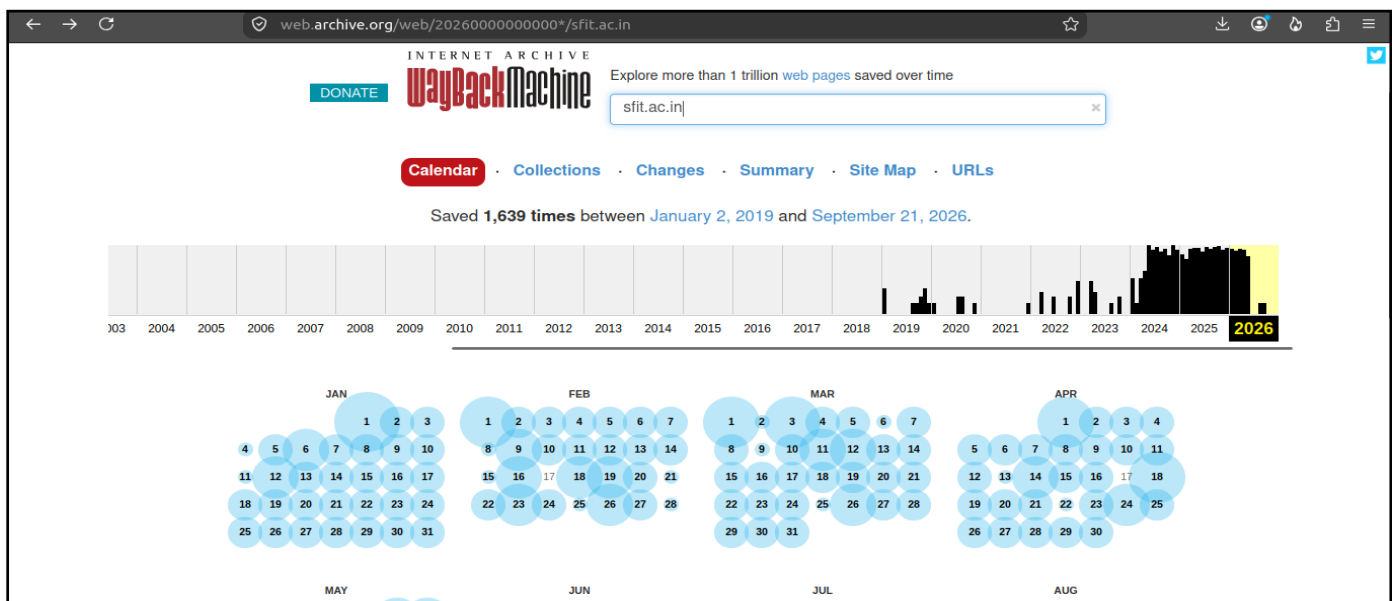
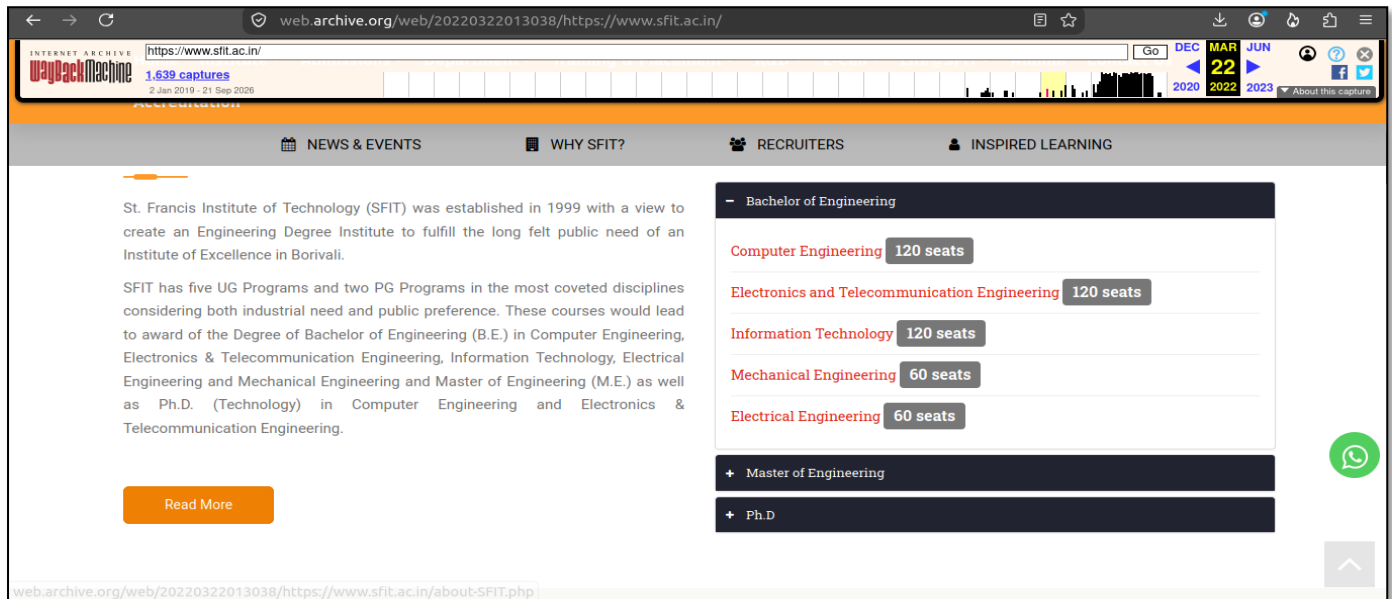
Non-authoritative answer:
Name:   sfit.ac.in
Address: 199.79.62.121
Name:   sfit.ac.in
Address: 64:ff9b::c74f:3e79
```

Observation:

The **nslookup sfit.ac.in** command queries the DNS server and displays the resolved IP address and related name-server information for the domain.

B. Passive Reconnaissance through publicly available tools

❖ archive.org:



Observation:

1. The **archive.org** shows the historical versions of the **sfit.ac.in** website captured over different dates.
2. The archive shows that the website has been saved many times, allowing previous **versions** of the website to be viewed.
3. The **calendar** indicates frequent captures in recent years, showing that changes to the website can be tracked over time.
4. The archived page shows older website content and layout, which can be compared with the present website for **passive reconnaissance**.

❖ Whois:

facebook.com Make offer Updated 3 hours ago

Domain Information

Domain:	facebook.com
Registered On:	1997-03-29
Expires On:	2034-03-30
Updated On:	2025-04-23
Status:	client delete prohibited client transfer prohibited client update prohibited server delete prohibited server transfer prohibited server update prohibited
Name Servers:	a.ns.facebook.com b.ns.facebook.com c.ns.facebook.com d.ns.facebook.com

354 GB Volume

Registrar Information

Registrar:	RegistrarSafe, LLC
IANA ID:	3237
Abuse Email:	abusecomplaints@registrarsafe.com
Abuse Phone:	+1-650-308-7004

Name	Organization
Domain Admin	Meta Platforms, Inc.
Address	
1601 Willow Rd	
Menlo Park, CA, 94025	
US	

Target Domain/URL/Website for whois : facebook.com			
Registrar:	RegistrarSafe, LLC	Registration Expiry date:	2034-03-30
Registration Update date:	2025-04-23	Name Servers:	a.ns.facebook.com, b.ns.facebook.com, c.ns.facebook.com, d.ns.facebook.com
Registrant Organization	Meta Platforms, Inc.	Registrant City:	Menlo Park

❖ Netcraft:

Network	
Site	http://microsoft.com 
Netblock Owner	Microsoft Corporation
Hosting company	Microsoft Corporation
Hosting country	 US
IPv4 address	150.171.109.241 (VirusTotal )
IPv4 autonomous systems	AS8075 
IPv6 address	2603:1061:14:f4:0:0:1
IPv6 autonomous systems	AS8075 
Reverse DNS	Unknown

Validity period	From Jan 22 2026 to Jan 17 2027 (11 months, 3 weeks, 5 days)
Matches hostname	Yes
Server	Not Present
Public key algorithm	rsaEncryption
Protocol version	TLSv1.3
Public key length	2048
Certificate check	ok
Signature algorithm	sha384WithRSAEncryption
Serial number	0x43000253929e1c999055f04653000000025392

Perfect Forward Secrecy	Yes
Supported TLS Extensions	RFC8446 supported versions, RFC8446 key share, RFC4366 server name, RFC4492 elliptic curves, RFC7301 application-layer protocol negotiation, RFC4366 status request
Application-Layer Protocol Negotiation	h2
Next Protocol Negotiation	Not Present
Issuing organisation	Microsoft Corporation
Issuer common name	Microsoft TLS G2 RSA CA OCSP 04
Issuer unit	Not Present

Issuer location	Not Present
Issuer country	 US
Issuer state	Not Present
Certificate Revocation Lists	http://www.microsoft.com/pkiops/crl/partition/Microsoft%20TLS%20G2%20RSA%20CA%20OCSP%2004_Partition00093.crl http://crl2.microsoft.com/pkiops/crl/partition/Microsoft%20TLS%20G2%20RSA%20CA%20OCSP%2004_Partition00093.crl
Certificate Hash	M+8qWEJJXsIHK2NfD5IfnpZub1c
Public Key Hash	dfd1a06be3dbe1fea78817bd8a9afdab57829819abf4c2bae78fdb34daae6193
OCSP servers	http://oneocsp.microsoft.com/ocsp
OCSP stapling response	Certificate valid
OCSP data generated	Sep 28 04:41:16 2026 GMT

Site Technology (fetched 12 days ago)

Server-Side

Includes all the main technologies that Netcraft detects as running on the server such as PHP.

Technology	Description
Using ASP.NET 🔗	ASP.NET is running on the server
SSL 🔗	A cryptographic protocol providing communication security over the Internet

Client-Side

Includes all the main technologies that run on the browser (such as JavaScript and Adobe Flash).

Technology	Description
Asynchronous Javascript	No description
JavaScript 🔗	Widely-supported programming language commonly used to power client-side dynamic content on websites

Target Domain/URL/Website for netcraft : **microsoft.com**

IPv4 address:	150.171.109.241	SSL/TLS certificate Issuing organization	Microsoft Corporation
Certificate Validity period:	22 Jan 2026 to 17 Jan 2027	Public key algorithm:	rsaEncryption
Public key length:	2048 bits	Certificate Hash:	M+8qWEJXslHK2NfDSIfnpZub1c
Signature algorithm:	sha384WithRSAEncryption	Public Key Hash:	dfd1a06be3dbe1fea78817bd8a9afdab57829819abf4c2bae78fdb34daae6193
Server-Side site technology:	ASP.NET, SSL	Client-Side site technology:	Asynchronous JavaScript, JavaScript